

SOP-MNT-009: Pump Vibration Monitoring

Standard Operating Procedure: SOP-MNT-009. Title: Centrifugal Pump Vibration Monitoring and Response.

Applicable Equipment: All centrifugal pumps rated above 50 HP, including Pump-101, Pump-102, Pump-105.

Regulatory Basis: OISD-118 (Pressure Vessel and Rotating Equipment Safety), internal QMS-SOP-014.

Procedure: 1) Record vibration readings weekly using handheld vibration analyzer. 2) Alarm threshold is 7.1 mm/s RMS per OEM specification. Trip threshold is 11.0 mm/s RMS.

3) If reading exceeds alarm threshold, raise a Work Order within 24 hours and notify the shift supervisor. 4) If reading exceeds trip threshold, equipment must be shut down immediately and isolated per LOTO procedure.

5) Standard lubrication interval is 90 days unless equipment-specific history (as recorded in the CMMS) indicates a shorter interval is required, in which case the shorter interval takes precedence.

Responsible roles: Maintenance Technician (readings), Shift Supervisor (Work Order approval), Senior QA Engineer (compliance sign-off during audits).